

Ex.No.7: Use AFLogical OSE to Extract Data from an Android Device

Aim

To extract logical data such as contacts, SMS, and call logs from an Android device using AFLogical OSE and analyze the collected evidence for forensic purposes.

STEP 1 — Prepare the Required Files

Files Required

- Android Platform Tools (ADB)
- AFLogical OSE APK
- Google USB Driver for Windows
- Android device
- USB data cable

Procedure

1. Create a main folder for the experiment:

C:\ForensicLab

2. Create the following folders:

C:\ForensicLab\platform-tools\

C:\ForensicLab\aflogical-ose\

C:\ForensicLab\output\

3. Place the downloaded files in their respective folders.
4. If the AFLogical OSE APK is not available, use the appropriate AFLogical OSE source or a forensic environment such as Santoku Linux.

STEP 2 — Add Platform Tools to PATH

Purpose

To allow ADB commands to be executed directly from Command Prompt.

Procedure

1. Open:

Control Panel → System → Advanced system settings → Environment Variables

2. Under **User Variables**, select **Path → Edit → New**.

3. Add the following platform-tools location:

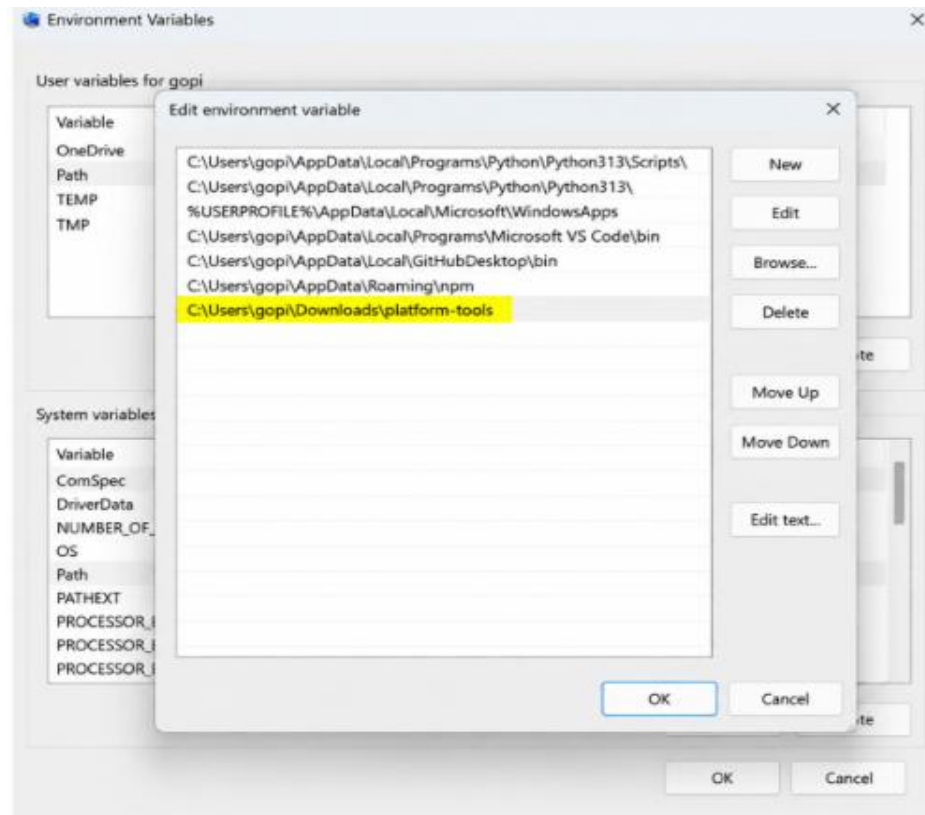
C:\Users\ASUS\Downloads\platform-tools

4. Click **OK** to save the changes.

Verification

Open a new Command Prompt and execute:

adb version



STEP 3 — Install Google USB Driver

Purpose To enable the Windows computer to communicate with the Android device through ADB.

Procedure

1. Connect the Android phone to the computer using a USB data cable.
2. Open Device Manager.
3. Locate the connected Android device.
4. Right-click the device and select Update Driver.
5. Select Browse my computer for drivers.
6. Select the Google USB Driver folder.
7. Click Next and complete the installation. Verification Open Command Prompt and execute:

adb devices

```
Command Prompt
Microsoft Windows [Version 10.0.26200.6901]
(c) Microsoft Corporation. All rights reserved.

C:\Users\gopi\Desktop\df\ForensicLab\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 36.0.0-13206524
Installed as C:\Users\gopi\Desktop\df\ForensicLab\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\gopi\Desktop\df\ForensicLab\platform-tools>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
10BD541MFE001FN unauthorized

C:\Users\gopi\Desktop\df\ForensicLab\platform-tools>|
```

STEP 4 — Enable Developer Options on the Phone

Procedure

1. Open Settings on the Android phone.
2. Go to About phone.
3. Tap Build number seven times.
4. Return to Settings and open Developer options.
5. Enable:
 - USB debugging
 - Install via USB, if available.

STEP 5 — Connect the Phone and Check ADB Connection

Purpose To verify communication between the forensic workstation and the Android device.

Procedure

1. Connect the phone using a USB data cable.
2. Unlock the phone.
3. Open Command Prompt or PowerShell.
4. Execute:

adb devices

5. If the phone displays an authorization request, select Allow USB debugging.
6. Run the command again:

adb devices

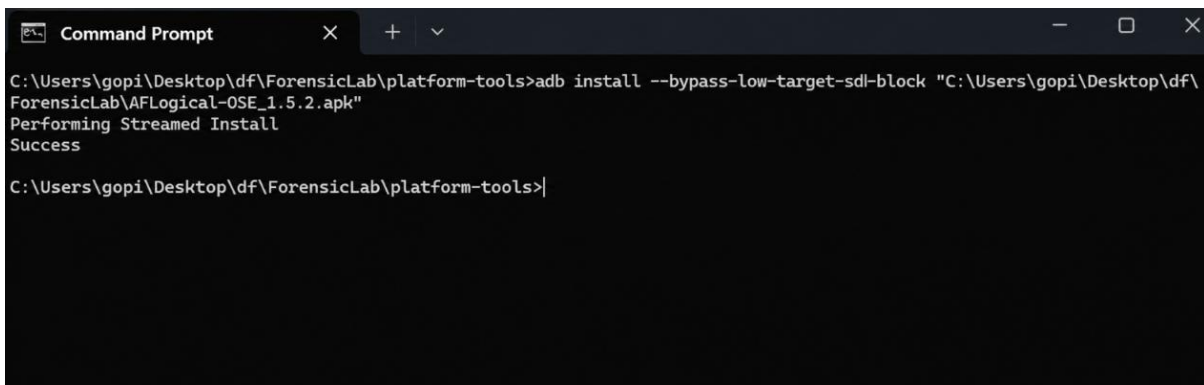
STEP 6 — Install AFLogical OSE on the Phone

Purpose To install the forensic extraction application on the Android device. Procedure

1. Ensure that the AFLogical OSE APK is available in: C:\ForensicLab\aflogical-ose\AFLogical-OSE.apk
2. Open Command Prompt.
3. Execute:

```
adb install "C:\ForensicLab\aflogical-ose\AFLogical-OSE.apk"
```

4. Wait for the installation to complete.
5. The following message indicates successful installation:



```
Command Prompt
C:\Users\gopi\Desktop\df\ForensicLab\platform-tools>adb install --bypass-low-target-sdl-block "C:\Users\gopi\Desktop\df\ForensicLab\AFLogical-OSE_1.5.2.apk"
Performing Streamed Install
Success
C:\Users\gopi\Desktop\df\ForensicLab\platform-tools>|
```

6. Check the Android device and verify that AFLogical OSE is installed.

STEP 7 — Run AFLogical OSE and Extract Data

Purpose To perform logical extraction of supported data from the Android device. Procedure

1. Open the AFLogical OSE application on the Android device.
2. Grant the required permissions.
3. Select the available data categories for extraction, such as:
 - Contacts
 - SMS
 - Call Logs
 - MMS
 - MMS Parts
4. Start the extraction process.
5. Wait until the extraction is completed. Verify the Extracted Data The forensic output can be checked using:

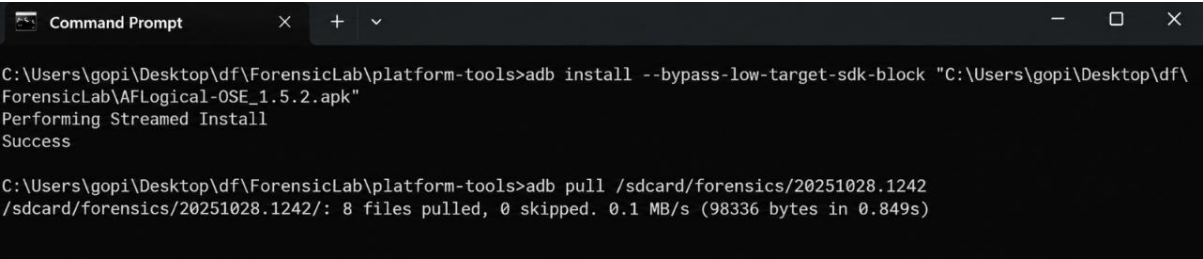
```
adb shell ls /sdcard/forensics
```

STEP 8 — Copy Extracted Data to the Computer

Purpose To transfer the extracted forensic data from the Android device to the computer for further examination. Procedure

1. Create the output folder:

mkdir C:\ForensicLab\output



2. Copy the extracted forensic data using: ``cmd adb pull /sdcard/forensics/ "C:\ForensicLab\output"
3. Wait for the transfer to complete.
4. Check the transferred files using: ``cmd dir "C:\ForensicLab\output"

Name	Date modified	Type	Size
CallLog Calls	23-09-2026 10:50	Microsoft Excel Com...	703 KB
Contacts Phones	23-09-2026 10:50	Microsoft Excel Com...	11 KB
e024862b-b46e-4548-8b59-cb868aa86344	23-09-2026 10:50	JPG File	179 KB
IMG_20231127_104636_01	23-09-2026 10:50	JPG File	137 KB
info	23-09-2026 10:50	Microsoft Edge HTM...	328 KB
MMS	23-09-2026 10:50	Microsoft Excel Com...	76 KB
MMSParts	23-09-2026 10:50	Microsoft Excel Com...	54 KB
SMS	23-09-2026 10:50	Microsoft Excel Com...	220 KB

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STEP 9 — Verify Integrity Using SHA-256

Purpose To verify the integrity of the extracted forensic evidence. Windows PowerShell Execute:

Get-FileHash "C:\ForensicLab\output<extracted-file>" -Algorithm SHA256

Record the generated SHA-256 hash value in the forensic report. Linux/macOS sha256sum

STEP 10 — Clean Up

After completing the extraction:

1. Close the AFLogical OSE application.
2. Uninstall AFLogical OSE from the Android device if required.
3. Disconnect the Android device safely.
4. Preserve the extracted forensic files and recorded hash values for documentation.

Result

The Android device was successfully connected to the forensic workstation using ADB. AFLogical OSE was installed and used to perform logical extraction of the supported mobile data. The extracted forensic output was transferred to the computer and examined for available contacts, SMS, call logs, MMS, and related information. SHA-256 hashing was used to verify the integrity of the selected extracted evidence.

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	